Learning Distributed Coloring Rules: Information, Counterfactual Fidelity, and Frozen Transfer

Research collaboration
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Abstract

When does a learned local policy execute a distributed algorithm rather than merely solve its task? We study seven explicitly specified coloring teachers and three neural palette architectures, all with learned communication. Across heterogeneous protocols, all 21 teacher–architecture combinations have five frozen checkpoints that each pass 1,000 native functional tests. This finite coverage result combines twelve historical and nine new certifications; it is not universal correctness. Functional validity and teacher fidelity separate sharply. For the degree-three algebraic base-case teacher, E1 attains perfect native action agreement on development teacher states under three procedures, yet mean continuation accuracy is 22.6404% for passive, 9.2686% for matched, and 99.9932% for counterfactual supervision. The latter uses more labels, and most sufficient COMMIT input types recur from training. We interpret this as a paired procedure comparison, not label-matched causal or broad intervention-generalization evidence. A frozen transfer study reconciles 810,000 confirmatory executions, including 450,000 primary-policy executions, across degree, size and palette conditions. Fifteen checkpoint–condition cells fail; full-batch replays distinguish proposal collapse, failed scheduled repair, and incorrect confirmation. Conditional proofs delimit teacher safety, progress, information requirements and a finite-type MAX certificate. The evidence identifies what these systems learn, where their rules fail, and which stronger claims remain open.

1 Introduction

Learning a local algorithm raises a stronger question than terminal task accuracy: under what information, supervision and input changes does a network reproduce an algorithm's decisions? Distributed coloring is useful because validity is easy to check, while anonymous randomness, identifiers, algebraic reduction and speculative repair yield distinct local rules. The same final proper coloring can conceal different decisions, trajectories, costs and probability laws.

We distinguish functional validity, action fidelity, transition fidelity, execution-cost fidelity, and equality of complete output laws. This general distinction is established in imitation learning and neural algorithmic reasoning. Our contribution is a concrete audited study within one distributed coloring framework. The central question is: *under which declared interfaces can local policies learn the specified rules, what do they reproduce, and where does frozen transfer fail?*

We document native functional coverage of seven variants by three communicating architectures with all 105 selected checkpoints identified. We expose a large separation between native agreement and typed continuation responses, auditing initialization, labels, shared heads and evaluation reuse. We reconcile frozen transfer and reproduce its failures, alongside conditional results separating safety, progress and representation. We claim neither a new fastest coloring algorithm nor a universally superior training method.

Relation to previous work. Neural graph coloring predates this study (Lemos et al., 2019), as do graph-algorithm execution with MAX aggregation (Veličković et al., 2020) and trace-based reasoning benchmarks (Veličković et al., 2022; Veličković and Blundell, 2021). DAgger addresses learner-induced state distributions (Ross et al., 2011); causal confusion in imitation already motivates interventions (de Haan et al., 2019). Our fixed-bank teacher continuations are not a new formulation of these general ideas. GNN/local-computation

connections and information restrictions are established (Loukas, 2020; Sato et al., 2019). Alignment and schedules affect extrapolation (Xu et al., 2021; Mathys et al., 2023); asynchronous robustness requires its own assumptions (Solodova et al., 2025), which we do not test. Recent coloring methods learn parallel orderings (Langedal and Manne, 2025), use recursive warm starts (Vanderbush and Weber, 2026), or learn contrastive coloring geometry with transfer (Le et al., 2026). Recent sufficient conditions for provable algorithm learning (Wittig et al., 2026) go beyond our finite extrapolation evidence. Our specific increment is the combination of scoped teacher interfaces, paired continuation diagnostics and replayable failure mechanisms, rather than neural coloring or size transfer themselves. Hint-ReLIC already regularizes algorithmic predictions across input augmentations preserving the next trace step (Bevilacqua et al., 2023); we instead acquire explicit teacher-continuation labels under alternative messages, whose targets can change. No-hint reasoning already separates terminal learning from trajectory imitation (Rodionov and Prokhorenkova, 2023). We claim neither algorithmic intervention nor that distinction as new. Appendix G compares candidate contributions with precedents.

2 Computational Model and Problem Definition

We use finite simple undirected labeled graphs $G = (V, E)$, $n = |V|$, $d(v) = |N(v)|$ and maximum degree Δ. Native training uses $\Delta \leq 3$ and $C_v = \{0, 1, 2, 3\}$. Transfer uses either the global list $C_v = \{0, \ldots, \Delta\}$ or local initial segments $C_v = \{0, \ldots, d(v)\}$. These are not arbitrary list-coloring instances. Partial colors satisfy $x_v \in \{-1\} \cup C_v$, where -1 is pending. Availability is $\mathcal{A}_v = C_v \setminus \{x_u : u \in N(v), x_u \neq -1\}$.

A teacher is a specified state machine with observations, private inputs, phases and an interpreter. The learned policy receives declared observations and emits decisions; the interpreter applies them. Anonymous trials use two neighbor exchanges; ColorTrials communicates availability-dependent preferences in the second. JP and Greedy need one exchange mathematically, but their neural decomposition retains two. GM has supplied blocks, barriers and a serial repair queue; BE supplies algebraic predicates and a reduction clock. These are distinct computational regimes, not equal-cost anonymous LOCAL algorithms. Message length and palette computation grow with palette size; we make no CONGEST complexity claim. Recorded native steps count state-machine applications. BE has two point-reduction steps and thirteen KW steps; neural point decisions use the first exchange, whereas KW evaluates the second stage as well. Fifteen native steps are not a fifteen-round learned communication complexity bound.

Mathematical random statements use fresh independent ideal draws. Implementation uses namespaced PCG64 streams and float32 observations. Required IDs are fixed distinct integers in $\{0, \ldots, 65535\}$. The implementation restricts $1 \leq n \leq 65536$, including anonymous experiments for convenience; this is not an anonymous-algorithm size lower bound.

Evaluation objects. Functional success means a complete proper in-list coloring within the protocol horizon and satisfaction of checked native invariants. A cutoff is a benchmark failure, not proof of nontermination. Action fidelity compares common states/private inputs. Transition fidelity compares updated states; different action codes need not yield different transitions. Continuation fidelity evaluates a declared oracle $O_T(S, a)$ after a typed message or color-query intervention. Such messages need not be sampled by the native teacher. Execution costs use teacher-specific units. Distributional fidelity concerns the full terminal vector and status on a fixed labeled graph. Terminal validity alone does not imply any of the other properties.

3 Teachers and Neural Architectures

Names identify the implemented variants. Bibliographic names identify source families, not guarantees imported without checking the variant. Appendix A specifies rules and conditional guarantees.

Shared communication. E1 NORMALIZED uses node/color MLPs, learned availability, private-input participation, a normalized ordered palette-prefix scorer and learned COMMIT after a second MAX exchange. DeepSets replaces mainly the palette scorer by shared color features and mean palette context. LocalPaletteTransformer uses two-head attention inside each node's palette. Both retain learned graph

Teacher	Exact rule	Supplied information/computation
Luby	Participate with probability 1/2, propose uniformly available, veto equal active offers	Pending/fixed colors, two private draws; anonymous
Johansson variant	Always active, uniform available proposal, equal-offer veto	Anonymous observations; no imported triangle-free theorem
ColorTrials	Minimum strictly winning available color, else wait	Four private preferences in $\{0, \dots, 3\}$ natively; two exchanges
JP	Pending strict ID maximum writes first-fit	Coherent relative ID comparisons; fixed IDs
BE_BASE_D3	Two polynomial reductions, thirteen KW reductions	Bounded IDs, fields $13, 7$, point/option equalities, bound/target and phase interpreter
Speculative GM	Four-block first-fit, same-slot marks, serial repair	IDs, slots, permissions, phases, queue built from predicted marks
Greedy	First-fit in increasing-ID permutation	One externally scheduled write permission per step

Table 1: Native teachers and distinct information regimes. ColorTrials is the repeated strict-preference subroutine; GM temporary conflicts are legal.

communication. Functional palette positions are supplied; color-renaming equivariance is not assumed. Hidden/message widths are 64/32. Canonical E1/DeepSets/Transformer have 16,035/17,907/16,995 parameters. Native residual adapters give 19,396/21,268/20,356 where required.

Learned components include participation, proposal scores and confirmation, BE point/option decisions, and GM marks. Polynomial evaluation, schedule construction, decoding, list-membership masks and fixed-order argmax are supplied. There is no exact availability mask, winning flag, eligibility decision or collision veto in neural inference. GM's queue follows predicted marks; an exact teacher never repairs the student's rollout.

Variable-degree representation. The historical serializer fixes four colors and three neighbor positions. Frozen transfer evaluates the same shared tensors with a sparse structural adapter. Native parity controls cover 21,630 decisions for 150 extensible selected/control checkpoints: no hard-action differences were observed, maximum absolute logit discrepancy 1.526×10^{-5}. Separately, 1,170 native executions cover all 195 original selected/control checkpoints. These are finite checks, not bitwise-equivalence proofs for every kernel. ColorTrials uses preference/Δ in transfer, agreeing with preference/3 natively; the out-of-domain scaling choice matters. For native-context teachers, participation averages first-stage features over each receiver's valid palette slots in the structural adapter. Fixed tensors therefore still use an explicitly adapted computation outside the historical serializer; palette membership is not the only adaptation.

4 Learning from Teachers

Native training samples sizes 8, 12, 16, 24 at degree at most three and global $K = 4$. Historical banks use six graph families; new banks add edgeless, mixed-component and clique/isolate examples. States, observations and optimization are part of each cohort's identity.

Let P, C denote participation/candidate messages. Passive training supervises native components on teacher prefixes. Its COMMIT loss evaluates the model's current messages against a stored native label, which may be inconsistent with the correct continuation for those messages. Matched training evaluates stored teacher P, C with their concordant native label, without acquiring new labels. Counterfactual training uses the same native bank plus four stored COMMIT labels per color root: it cycles the queried root through four colors with teacher-specific message interventions and a silence case. After matched warmup, its loss is

$$L = L_P + L_C + \tfrac{1}{2} L_{\text{native,matched}} + \tfrac{1}{2} L_{\text{continuation}}.$$

LINIAL point and GM DETECT phases retain their native losses. These fixed teacher-state banks differ from collecting states visited by the learner.

Paired arms share architecture, seed, initialization, native data, minibatch order, Adam learning rate 10^{-3} and batch size 64; at a fixed milestone they share updates. CF adds labels and changes the confirmation loss distribution. Thus matched versus CF does not identify an intervention-only effect or efficiency at equal total label budget. Acquired component labels, model access, presentations, updates, bytes and CPU time are separate costs. Replaying a stored label is optimization, not new supervision. Shared banks are charged once for physical acquisition, separately for each model's access.

Admissibility. Anonymous queries require proper partial prefixes. JP signs derive from a coherent fixed ID order. BE states come from legal polynomial/KW replay; derived predicates cannot be edited independently. GM requires legal speculative history and schedule. A typed rooted continuation need not form a jointly consistent graph-wide intervention. We claim neither arbitrary state robustness nor joint intervention fidelity from root queries.

Selection and replication. The historical sample-budget study has seven budgets 10^3–10^6, 500 updates and five seeds: 735 policies. New procedures record 500, 1,500 and 4,000-update milestones. A prospective amendment expands development from 60 to 300 runs before new confirmation. Five candidate hashes and a final namespace are frozen before a once-only 1,000-run battery per policy. Failed candidates remain; subsequent follow-ups use new namespaces. Local registration is supported by Git/artifact provenance, not independently timestamped external preregistration. Appendix C supplies the budget/provenance table and costs.

5 Functional Learnability

The native result is **21/21 combinations under their respective protocols**: five selected checkpoints per combination, each with 1,000/1,000 proper complete invariant-valid executions. Twelve (Luby, Johansson, JP, Greedy across three cores) are historical passive packages, seeds 83101–83105. Nine (ColorTrials, BE, GM across three cores) are new CF packages, seeds 94101–94105. The representative BE/CT weights use 4,000 updates and GM 500. Matched controls also close all nine; CF is not shown necessary for native functional success.

Teacher	E1	DeepSets	Transformer
Luby	H: 5/5	H: 5/5	H: 5/5
Johansson	H: 5/5	H: 5/5	H: 5/5
ColorTrials	N: 5/5	N: 5/5	N: 5/5
JP	H: 5/5	H: 5/5	H: 5/5
BE base	N: 5/5	N: 5/5	N: 5/5
GM	N: 5/5	N: 5/5	N: 5/5
Greedy	H: 5/5	H: 5/5	H: 5/5

Table 2: Native coverage: each passing checkpoint has 1,000/1,000 successes. H: historical passive; N: new CF. These protocols do not form a homogeneous architecture-ranking experiment.

Final tests use sizes 32, 48, 64; family weights differ: seven historical strata versus ten new strata of 100 runs. The 105 selected policies account for 105,000 successful observations, not a success probability of an average policy. Five training seeds are the treatment replication units. Under independent Bernoulli trials with a common probability, 1,000/1,000 gives pointwise two-sided 95% Clopper–Pearson lower limit $0.025^{1/1000} = 0.996318$; the one-sided limit is $0.05^{1/1000} = 0.997009$. For fixed independent nonidentical strata, AM–GM makes the zero-failure formula conservative for the weighted average, not each stratum. No simultaneous all-policy guarantee or universal correctness follows. A selected successful budget is not a theoretical minimum. The run-level interpretation also assumes semantically neutral batching. Real-arithmetic message separation supports that premise; universal numerical neutrality across floating-point batch contexts has not been certified.

Preserved negative results. The historical study's nine noncertifications under 500 updates are not impossibility results. The new study retains 21 failed executions among 195,000 final observations across 39 packages and 170 distinct checkpoints; some weights receive new batteries. All failures remain separate from the selected all-pass set. Independent fixed-bank E1 controls compare 500 versus 1,500 updates along the same seed/checkpoint lineage, holding labels fixed and using fresh final tapes. This is a more narrowly identified optimizer contrast than matched versus CF; it does not establish a CF effect or monotone benefit from more updates (Appendix C).

6 Counterfactual Fidelity

We reconstruct the existing paired 4,000-update diagnostics on 60 exact teacher trajectories at sizes 28 and 36 per seed. Their graph/tape namespaces are disjoint from training. They belong to the repeatedly used development split, not an untouched final fidelity battery. BE/E1 gives 28,800/28,800 native permitted-root action agreements in every arm and seed, but very different continuation responses: The paired initial hashes,

Procedure	Labels	Mean acc. (%)	Seed range (%)	Mean BA (%)
Passive	3,000	22.6404	9.1607–42.3297	60.1413
Matched	3,000	9.2686	9.0905–9.3970	53.2536
Counterfactual	6,776	99.9932	99.9780–100.0000	99.9965
Always reject	–	97.0465	96.8800–97.2226	50.0000
Eligibility only	–	98.5990	98.5146–98.6208	99.2781

Table 3: BE/E1, 4,000 updates. Each seed has 99,840 continuation queries. Means describe five paired training seeds; 499,200 queries per arm are correlated within states/trajectories, not independent training replicas. BA averages sensitivity and specificity within seed, then over seeds. Eligibility-only accepts iff the root is KW-eligible and participating; these semantic baselines are diagnostics, not trained competitors.

native arrays, visit counts and final minibatch RNG states agree. CF adds 3,776 labels, changes COMMIT contexts and splits its confirmation loss. This supports a difference between the full procedures, not an intervention-only effect, sample efficiency, or universal functional improvement. Only 2.7774–3.1200% of queries are positive. Every arm has zero false negatives; CF has 34 false positives, all on eligible but silent roots. Most of the raw gap concerns rejection of ineligible KW continuations. Among eligible active queries (4,150–4,598 occurrences per seed), CF is 100% correct in all seeds, whereas passive and matched accept every option, including occupied ones: mean accuracy 67.8117%, BA 50%. Thus the effect also includes learning to reject occupied options, not only a redundant eligibility gate. Always rejecting is accurate overall but destroys progress.

Novelty of the evaluation queries. Reconstructing the original query RNG reproduces every numerator. Only 91 of 499,200 occurrences exactly match a complete serialized training query. However, BE COMMIT discards many serialized fields. Its queried-slot logit depends on phase, permission, signed KW offset, root participation, and the MAX-set of occupied/participating/offering neighbor types. Between 99.277% and 99.540% of diagnostic occurrences have one of these sufficient input types already in training. The four queried colors and intervention families are not held out. Among the 549, 459, 717, 522, 722 occurrences with unseen sufficient types, CF correctness is 549, 437, 717, 510, 722, versus matched 158, 342, 229, 182, 246. This posthoc restricted result preserves a seed-wise advantage. Its correlated small subsets and development reuse do not support broad new-intervention-distribution generalization. Appendix D defines the sufficient-input key and gives per-seed numerators. These unseen-type subsets include 52,108,90,56,60 positives and both classes in every seed. All three arms accept all positives; CF's advantage is again specificity on negative continuations. The mean unseen-subset BA is 99.1157% CF versus 66.5299% matched and 50% always-reject.

Other teachers and dependent architecture comparisons. At the same milestone GM/E1 has mean continuation accuracy 75.3632% for passive/matched versus 100% CF; CT has 96.6570%, 97.2439%,

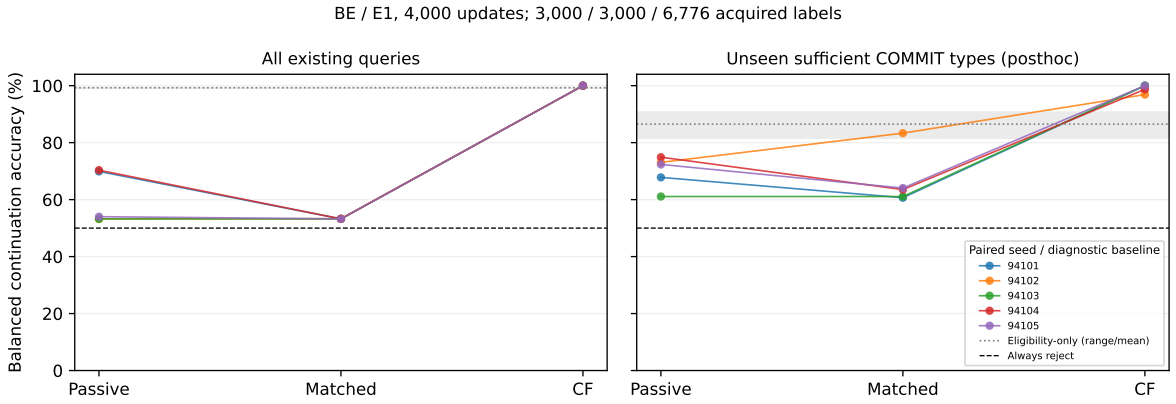


Figure 1: BE/E1 balanced accuracy across paired seeds, with constant-reject and eligibility-only diagnostic baselines. Right: posthoc unseen sufficient COMMIT types. Curves connect treatments within each paired seed; label budgets differ by arm.

99.7452%. Their denominators vary by seed. These are supporting diagnostics. Paired BE/GM CF COMMIT tensors are identical across the three palette cores at audited milestones: fixed-proposal second-stage losses do not involve their palette scorer. These are five paired confirmation replicas per milestone, not fifteen independent architecture replicas. CT couples second-stage gradients to proposal scoring and lacks that tensor identity.

The scoped answer is positive: this CF procedure better reproduces these typed teacher continuations, including some unseen local types, even when passive native agreement is already perfect. It is not always necessary, universally functionally superior, or shown more label efficient. No new training campaign is used to rescue those stronger claims.

7 Information and Identifiers

Consider an initial two-node JP graph with identical anonymous observations and fixed private inputs. Swapping the distinct IDs changes the required root action from write to wait without changing the anonymous observation.

Proposition 1 (Observation indistinguishability)**.** *If two admissible contexts have the same observation and different deterministic target actions, no deterministic function of that observation is correct on both. A randomized function cannot be almost surely correct on both.*

Proof. Equal inputs give equal outputs or equal conditional output laws. Different point-mass targets cannot both be realized. For equiprobable binary contexts, average correctness is at most 1/2. □

This elementary interface obstruction is not a new distributed lower bound. It requires indistinguishability, not numerical closeness, and does not preclude solving coloring by a different rule. Historical integrated JP/GM already had numeric ID differences; their failures were not ID blindness.

JP representation	94101	94102	94103	94104	94105
Without ids	6/60	6/60	6/60	6/60	6/60
Numeric ids	13/60	11/60	6/60	26/60	15/60
Numeric differences	16/60	15/60	19/60	31/60	20/60
Relative comparisons	60/60	60/60	60/60	60/60	60/60

Table 4: Exploratory E1 JP control: same dimensions, initialization, 3,000 labels and 500 updates. Entries are development successes out of 60, not final 1,000-run certifications.

Relative comparisons achieve 60/60 in all five seeds. This supports representation sensitivity under this budget. Numeric IDs/differences retain order information, so finite failures do not prove inability to learn it. Signs remove the need to resolve gaps as small as 1/65536; they must arise from one coherent order. BE uses actual ID arithmetic: increasing numeric transformations need not preserve its polynomial trajectory.

8 Generalization

Teacher extensions. The six non-BE rules admit the parameterized definitions in Appendix A. JP's historical cap becomes n, preserving its priority rule. GM retains four blocks/detection/serial repair; Greedy retains its schedule. Palette lists extend the domain. CT also changes its preference alphabet to $\{0, \ldots, \Delta\}$; it is a declared parameterized variant, not identical fixed-$K4$ private inputs. BE is excluded because fixed fields and four-option blocks lack a higher-degree existence proof. Changing those constants would define another mathematical variant and interface.

Confirmatory design. After a separate pilot, weights, conditions, namespaces and horizons are frozen. All 150,000 exact-reference executions pass before students run. Ninety primary extensible checkpoints receive 1,000 executions at each of five conditions: global $n = 64, \Delta = 3, 4, 10$; local $n = 64, \Delta = 4$; global $n = 256, \Delta = 4$. Sixty native-certified matched/passive controls and thirty pre-existing CT 4,000-update controls receive the four $n = 64$ conditions. The thirty are not assumed certified. Weights stay frozen, inference uses float32 CPU. Each graph has maximum degree exactly requested. Conditions use a uniform mixture of regular, capped random, bipartite, tree, irregular, hub, disconnected, isolate-containing, heterogeneous and clique families. Capped random is not an Erdős–Rényi law. Instances/tapes are shared across policies, teachers and cores. There are 4,000 distinct labeled graphs, 20,000 tape seeds and 25,000 case–palette pairs in the 810,000 policy observations. Observation, graph and training-replica counts differ.

Raw-color reconstruction gives **808,413/810,000** correct: 1,559 collisions and 28 cutoffs in fifteen of 810 cells. This is accounting across policies, not the success rate of a mean algorithm. Primary policies give **449,791/450,000**: 209 failures in two cells, with 90,000/90,000 degree-three controls. Secondary controls give 358,622/360,000. Pointwise checkpoint intervals describe the declared mixture, not 1,000 executions per graph family. Table 5 preserves seed ranges.

Teacher	Core	$64, 3, G$	$64, 4, G$	$64, 10, G$	$64, 4, L$	$256, 4, G$
Luby	E1	5/5	5/5	5/5	5/5	5/5
Luby	DeepSets	5/5	5/5	5/5	5/5	5/5
Luby	Transformer	5/5	5/5	5/5	5/5	5/5
Johansson	E1	5/5	5/5	5/5	5/5	5/5
Johansson	DeepSets	5/5	5/5	5/5	5/5	5/5
Johansson	Transformer	5/5	5/5	4/5 (972–1000)	5/5	5/5
ColorTrials	E1	5/5	5/5	5/5	5/5	5/5
ColorTrials	DeepSets	5/5	5/5	5/5	5/5	5/5
ColorTrials	Transformer	5/5	5/5	5/5	5/5	5/5
JP	E1	5/5	5/5	5/5	5/5	5/5
JP	DeepSets	5/5	5/5	5/5	5/5	5/5
JP	Transformer	5/5	5/5	5/5	5/5	5/5
GM	E1	5/5	5/5	5/5	5/5	5/5
GM	DeepSets	5/5	5/5	5/5	5/5	5/5
GM	Transformer	5/5	5/5	5/5	4/5 (819–1000)	5/5
Greedy	E1	5/5	5/5	5/5	5/5	5/5
Greedy	DeepSets	5/5	5/5	5/5	5/5	5/5
Greedy	Transformer	5/5	5/5	5/5	5/5	5/5

Table 5: Primary checkpoints with 1,000/1,000 successes out of five. Nonperfect cells also give the per-checkpoint success range. Columns are $(n, \Delta, \text{palette})$, G global and L local. Sixteen of eighteen combinations pass all selected conditions.

Failure mechanisms. Fresh replays preserve the original complete inference batch containing the first failure in each failed cell. All fifteen reproduce final colors, status and steps, avoiding batch-shape numerical changes to the witness. Johansson/Transformer 83101 has 28/1,000 degree-ten cutoffs, all on cliques. The first leaves two adjacent pending vertices with free colors $\{0, 2\}$. Both repeatedly propose color 2, correctly

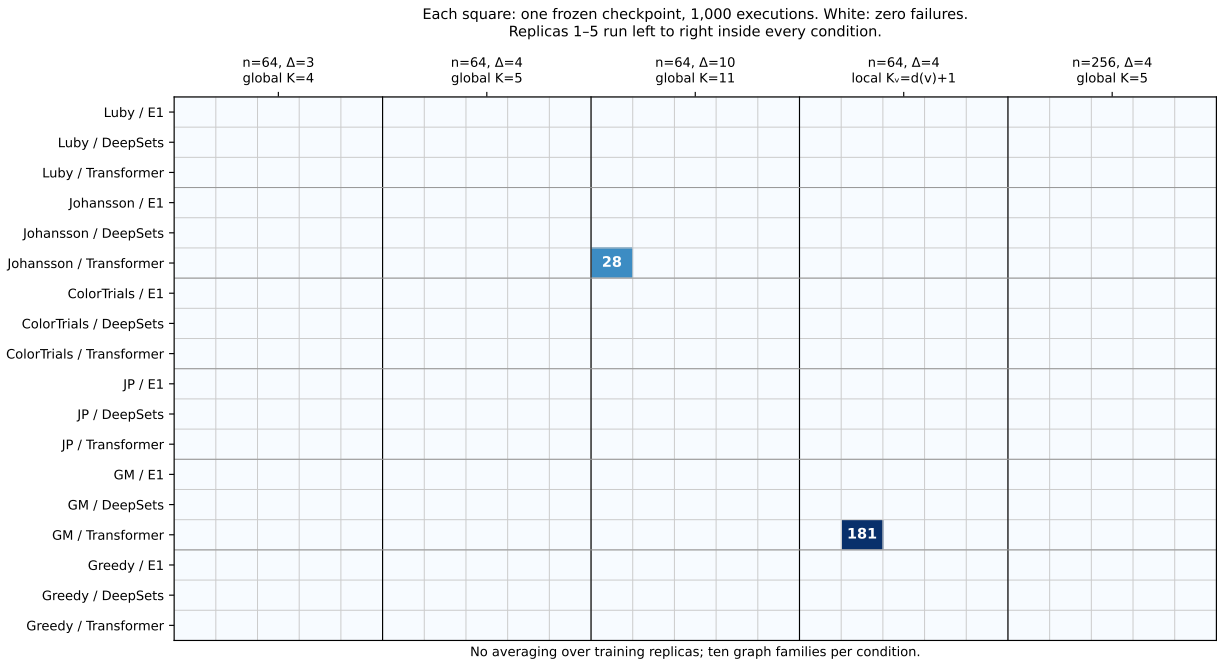


Figure 2: All primary frozen replicas. Each square represents 1,000 runs; numbers indicate failures, white squares zero failures. Palette size changes with degree; global success does not imply local-list success.

vetoed by COMMIT. In the preserved conditional probe each proposes 2 on all 8,192 new draws. This identifies missing observed proposal diversity in that state, not an incorrect veto or exactly zero mass for the other color.

GM/Transformer 94102 has 181/1,000 local degree-four failures in each of CF, matched and passive. In the first primary CF witness, exact assignment and detection precede a wrong repair proposal: correct rejection omits a scheduled repair and leaves a conflict. Matched/passive first witnesses include a forbidden write. Equal failure counts therefore do not imply identical local mechanisms. A separate exact-prefix control fixes $\Delta = 3$, IDs and endpoint colors but changes a degree-one endpoint's list from four to two; the Transformer misses the exact repair. This isolates palette-context sensitivity in that constructed state, not universal capacity versus optimization explanations.

Eleven other failed cells involve three matched E1 CT checkpoints, including degree-three failures and incorrect simultaneous confirmations. Higher degree cannot explain all of them. Alpha-threshold discrepancies are latent calibration diagnostics, not supervised availability errors or causes by themselves. The first temporal discrepancy is not automatically sufficient to explain a final failure. Appendix E lists every cell.

Exploratory scope. The separate pilot has 3,526/3,570 successes: two collisions, 42 CT resource cutoffs. It spans regular $n = 256$, degrees 2,3,4,5,10,20,50,100,200, sizes through 1,024 at degree 3 or 10, plus heterogeneous/local cases. A ten-second cap was applied only to unfinished dense CT cells after some longer degree-50 successes existed. These unequal resource horizons do not permit homogeneous success-rate ranking. One run per checkpoint/cell is exploratory. The later size pilot gives 240/240 for sixty Luby/Johansson/CT/JP checkpoints on regular $n = 2048, 4096$, degrees 3,10, one run per point. It contains four labeled graphs and twenty tapes, separately from confirmation. GM/Greedy size pilots stop at 1,024; BE stays outside this extension. Confirmation therefore reaches degree ten at size 64, and size 256 at degree four, not their full cross-product, degree 200, or size 4,096.

Scale witness and confounding. Degree changes global K and CT preferences simultaneously. A posthoc typed probe keeps an edge endpoint's degree one and local list $\{0, 1\}$ fixed, while a separate star

raises global Δ. Preferences Δ and $\Delta - 1$ make it the strict color-zero winner. All fifteen primary CF CT policies reject its queried COMMIT from degree ten in the main embedding. Legacy preference/3 units alter responses but do not fix all tested cases. Both encodings preserve order. This is scale sensitivity of fixed policies, not an information obstruction, a demonstrated rounding cause or a random functional failure rate. No degree-only causal effect independent of K and private-input scale is identified.

9 Discussion

These systems learn executable local modules within a supplied interpreter. Native all-pass packages demonstrate finite functional learnability, not identification of each teacher's entire algorithm. BE is informative because native agreement leaves continuation behavior underconstrained. The budget, overlap and shared-head audits limit its interpretation to the measured procedures, with posthoc evidence on some unseen local input types.

Safety and progress. Exact availability and equal-offer rejection preserve partial properness. An always-wait policy is safe forever. On a K-clique, all nodes proposing one color with independent participation $1/2$ progress with probability $K/2^K$ per round. Diffuse independent proposals permit progress bounds, but those uniform hypotheses are not established for these checkpoints. MAX can implement a presence predicate and a finite-type COMMIT certificate (Appendix B); it does not establish candidate diversity, termination or palette-context invariance. All 45 selected canonical transfer cores fail the sufficient all-types condition, despite successful rollouts; five different historical E1 weights pass conditional certificates. These cohorts cannot be conflated. Vertex equivariance assumes consistent relabeling of observations, IDs and schedules in real arithmetic; ordered color features do not imply color-name equivariance.

Output laws and limitations. Properness and one-node marginals do not determine joint coloring laws. Historical studies use complete vectors/statuses, run-level permutations, fixed graph/checkpoint blocks and explicit testing families. They are supporting evidence, separate from the 810,000 runs (Appendix F). Nonrejection is never equality; coupling and symmetry justify only stated bounds or special cases. Native cohorts differ in observations, supervision and budgets; combined coverage cannot rank architectures fairly. CF uses more labels, development states and familiar query templates. Shared heads limit independent architecture evidence. Pilot-selected conditions and paired graph/tape reuse limit extrapolation. Degree, palette and private-input scale are confounded. Failure witnesses do not settle every capacity/optimization cause. Teacher proofs use ideal randomness; hardware arithmetic and finite PRNGs need separate assumptions. Forty original Pedro weights remain unavailable; their historical traces are background, not a fully replayed or imputed cohort.

We freeze scope. A label-matched CF comparison with new final states, held-out intervention families and paired seeds would address a stronger causal claim, not a prerequisite for the procedure comparison here. Multigrade retraining, general BE, asynchrony and architecture search remain future work.

10 Conclusion

Seven specified teachers and three communicating architectures attain native coverage under finite heterogeneous protocols. Paired continuation diagnostics show that native agreement can coexist with different teacher responses; resource, overlap and shared-head audits restrict causal interpretation. Frozen transfer works widely in selected conditions and fails through identifiable proposal, repair and confirmation mechanisms. Conditional mathematical results do not automatically transfer to learned execution. These scoped findings explain what the systems reproduce and why successful coloring alone does not identify the learned distributed rule.

A Exact Rules and Conditional Guarantees

Assume finite simple graphs, proper partial states where required, declared lists of size at least $d(v) + 1$, coherent observations and the specified synchronous exchanges/schedules. Code domain guards and independent scalar tests complement these arguments.

Lemma 2 (Availability)**.** *For a pending vertex,* $|\mathcal{A}_v| \geq |C_v| - (d(v) - d_U(v)) \geq d_U(v) + 1$.

Proof. At most one distinct color per fixed neighbor is forbidden. Counting duplicates only weakens the bound; differing local lists cause no problem. □

Luby/Johansson. Proposals avoid fixed colors, and exact equal-active-offer rejection prevents simultaneous collisions. Permanent colors persist, giving safety by induction. Condition on history/state before new private inputs, not a snapshot already containing those inputs. If $q_v(c)$ is the proposal law, p_v its independent participation, with independent pairs across nodes, then

$$\Pr(v \text{ commits} \mid S) = p_v \sum_{c \in \mathcal{A}_v} q_v(c) \prod_{u \in U_v} (1 - p_u q_u(c)). \tag{1}$$

For uniform proposals and $p_u = 1/2$, the chord bound $\log(1 - x) \geq -2\log(2)x$ on $[0, 1/2]$ and Jensen yield $\frac{1}{2}2^{-d_U(v)/|\mathcal{A}_v|} \geq 1/4$. For always-active Johansson, each pending neighbor of v has at least two free colors, so $q_u(c) \leq 1/2$. The same argument yields $2^{-2d_U(v)/|\mathcal{A}_v|} \geq 1/4$. These prove the specified variants' bounds; they do not import a different algorithm's theorem (Luby, 1993; Johansson, 1999).

ColorTrials. Let $D \geq \Delta$ be the common preference bound: $D = 3$ natively, even on lower-degree graphs, and $D = \Delta$ in the extension. Pending nodes draw independent preferences for each color in $\{0, \ldots, D\}$. Only pending neighbors for which the color is available compete. The minimum strictly winning available color is chosen; otherwise wait. Adjacent vertices cannot both strictly win one color. For any $c \in \mathcal{A}_v$, the event of preference D at v and smaller preferences at all at most Δ competitors has probability at least $\rho_{\mathrm{CT}} = (D + 1)^{-1}(D/(D + 1))^{\Delta} > 0$ for $D > 0$. It guarantees some write at v, perhaps a lower winning color. Isolates at $\Delta = 0$ finish in one step. The common global preference alphabet matters; different local alphabets need another analysis. This is a repeated subroutine inspired by symmetry breaking (Schneider and Wattenhofer, 2010), not a full multi-phase complexity claim.

Random termination. If each pending vertex has conditional progress at least $\rho > 0$ at every reachable state, iterated conditioning gives $\Pr(T_v > t) \leq (1 - \rho)^t$ and $\Pr(T > t) \leq \min\{1, n(1 - \rho)^t\}$. Thus ideal exact rules terminate almost surely. Transfer uses at least 128 steps, enlarged for CT until this ideal tail bound is at most 10^{-9}. Neither arbitrary learned policies nor finite PRNG orbits inherit the result automatically.

JP and Greedy. JP's pending strict maxima of coherent distinct IDs form an independent set. First-fit exists by availability, and the remaining global maximum writes each step, giving safety and at most n steps. This synchronous variant is inspired by Jones and Plassmann (1993). Greedy permits exactly one vertex in increasing-ID order per step; first-fit avoids prior colors and completes in n writes. Global scheduling cost is supplied, not a free LOCAL operation.

Speculative GM. Split increasing-ID order into $\min(4, n)$ balanced blocks. ASSIGN visits one position per block simultaneously. A later assignment avoids all earlier neighbor colors, so only equal-slot vertices can conflict. DETECT marks the smaller-ID endpoint of each equal-color/equal-slot edge. Every conflict is covered; the unmarked subgraph is proper. Serial repair visits marked vertices once in increasing-ID order and avoids all current neighbor colors. The last repaired endpoint of any edge avoids the other; edges with no repaired endpoint were already proper. Correctness follows in $2\lceil n/\min(4, n)\rceil + m$ steps, $m \leq n$. Every scheduled write/mark must be correct. Missed learned repairs are not retried. This is the implemented four-block variant (Gebremedhin and Manne, 2000), including supplied barriers/queue, not a guarantee for arbitrary learned marks.

BE degree-three base case. Distinct IDs are initial proper auxiliary colors. Degree-at-most-four polynomials over $\mathbb{F}_{13}$ encode them. A different neighboring polynomial agrees at at most four points, so three neighbors block at most twelve and $13 > 3 \cdot 4$ leaves a point. Encode selected point/value below $13^2 = 169$. A second degree-at-most-two reduction over $\mathbb{F}_7$ has a point because $7 > 3 \cdot 2$, yielding bound 49. Different chosen points give different pairs; equal points have different unblocked values, proving properness (Linial, 1992). For bound $B > 4$ set $q = \lceil 4B/5 \rceil$. Colors below q remain; each high old color owns a disjoint four-option block in $[0, q)$, possible since $4(B - q) \leq q$. Three neighbors forbid at most three options. High–high neighbors have disjoint blocks; high–low collisions are excluded by free options. The sequence $49 \to 40 \to 32 \to 26 \to 21 \to 17 \to 14 \to 12 \to 10 \to 8 \to 7 \to 6 \to 5 \to 4$ has thirteen KW reductions (Kuhn and Wattenhofer, 2006). Decoding does not search for a good point/option. This is only the degree-three base construction associated with Barenboim and Elkin (2008), not the general recursion. At degree four, $13 > 4 \cdot 4$ fails; changing constants changes the variant.

B Conditional Representation and Probability Results

The following elementary arguments are not claimed new general GNN-learning theorems. Checkpoints satisfy their hypotheses only where verified.

Proposition 3 (Diffuse proposals)**.** *Assume a proper state, exact rejection, independent node-private draws, independence of participation/candidate within nodes,* $0 < p_{\min} \leq p_v \leq p_{\max} < 1$*, proposal support in* $\mathcal{A}_v$*, and* $\max_c q_v(c) \leq \beta/|\mathcal{A}_v|$*. Then* $\Pr(v \text{ commits} \mid S) \geq p_v(1 - p_{\max})^{\beta d_U(v)/|\mathcal{A}_v|} \geq p_{\min}(1 - p_{\max})^{\beta}$*. Only* v *needs the concentration bound for this individual-node statement.*

Proof. The log chord gives $\log(1 - x) \geq ax$ for $a = \log(1 - p_{\max})/p_{\max} < 0$, $0 \leq x \leq p_{\max}$. Write $\Lambda_c = \sum_{u \in U_v} p_u q_u(c)$. Equation (1) is at least $p_v \sum_c q_v(c) e^{a\Lambda_c} \geq p_v e^{a \sum_c q_v(c)\Lambda_c}$ by Jensen. The weighted load is at most $\beta p_{\max} d_U(v)/|\mathcal{A}_v|$; each neighbor has mass at most one on $\mathcal{A}_v$. Multiplication by $a < 0$ reverses that inequality, and availability gives the last bound. Isolates use the empty product. □

Corollary 4 (Diffuse submass)**.** *If the concentration bound holds only on* $S_v \subseteq \mathcal{A}_v$ *of mass* $\eta_v > 0$*, the lower bound is* $p_v \eta_v (1 - p_{\max})^{\beta d_U(v)/(\eta_v |\mathcal{A}_v|)}$*. Uniform* $\eta_v \geq \eta > 0$ *at every reachable state implies uniform progress* $p_{\min} \eta (1 - p_{\max})^{\beta/\eta}$*.*

Proof. Drop nonnegative terms outside S_v, renormalize by η_v, and repeat. $xb^{\beta/x}$ increases for $x > 0$, $0 < b < 1$, since its derivative is $b^{\beta/x}(1 - \beta \log b/x) > 0$. □

Counterexample 5 (Safety without useful progress)**.** *On a* K*-clique with* $K \geq 2$ *colors, deterministic color-zero proposals, participation* $1/2$ *and exact rejection progress only with exactly one active vertex: probability* $K/2^K$*, expected first-progress time* $2^K/K$*. Always-active vertices never progress. A shared uniform candidate draw has uniform individual marginals and the same slow law; independence in* (1) *cannot be replaced by marginal uniformity.*

Proposition 6 (Canonical finite-type MAX COMMIT)**.** *Assume legal canonical per-color inputs/gating, nonnegative shared second messages, componentwise MAX with zero empty aggregate, and exact proposed-slot gather. The pending active center's queried-slot vector is* $(1, 0, 1, 1)$*; neighbor vectors* $(pending, owns_c, participates, offers_c)$ *are among*

$$(0,0,0,0), (0,1,0,0), (1,0,0,0), (1,0,0,1), (1,0,1,0), (1,0,1,1).$$

If all 64 type-subset COMMIT signs accept precisely subsets without the fixed-color or active-equal-offer conflict types, the rule is exact at any legal degree/palette in real arithmetic and preserves properness.

Proof. MAX is associative, commutative and idempotent; multiplicities disappear. Nonnegative messages admit zero identity. Gathering the proposed slot removes other slots. The 64 subsets therefore enumerate all aggregates. Excluding the two conflict types prevents old/simultaneous collisions. Induction gives safety, with no progress conclusion. □

This does not cover continuous preference or native residual contexts, multiple offers, SUM or MEAN. Outward-rounded float64 intervals enclose float32-stored weights as real coefficients. A separate mode adds γ_{m+2} float32 affine error and a subnormal allowance, conditional on IEEE rounding, no overflow and no TF32/lower precision. It is not a proof for arbitrary GPU kernels. Five historical E1 weights pass; all 45 selected Luby/Johansson/Greedy transfer cores fail the sufficient all-types condition. Failure of a sufficient certificate alone does not prove a reachable failure.

Proposition 7 (MEAN dilution; SUM construction)**.** *Fix degree/palette-independent messages m_o, m_f and a continuous classifier F for a fixed active center. If $F(m_o) > 0$, a MEAN interface accepts a conflicting star with sufficiently many ordinary neighbors. A SUM interface can represent exact conflict presence in real arithmetic.*

Proof. A star with d ordinary leaves and one fixed neighbor owning the candidate has mean $(dm_o + m_f)/(d+1) \to m_o$. Continuity and strict acceptance imply false acceptance at some finite d. If F is L-Lipschitz, it suffices that $d+1 > L\|m_f - m_o\|_\infty / F(m_o)$. For SUM, use the legal-type conflict bit $b = o + \mathrm{ReLU}(p + a - 1)$ and head $1/2 - \sum_u b_u$. □

The MEAN statement assumes no separate degree/palette input and says nothing about reachability of the constructed state. SUM's observed failure is not impossibility of representing this predicate. An order-free equivariant slot scorer with identical available-slot inputs and invariant pooled context gives tied scores for fixed private input; it cannot select distinct ordered ranks without order features or external tie-breaking. Main scorers have positions, so this obstruction does not explain their errors automatically.

C Training, Certification and Historical Provenance

Table 6: Selected native packages: actual acquired labels per seed (ranges where necessary), updates and total parameters. Every row has five policies and 1,000/1,000 successes each. Historical seeds 83101–83105, new 94101–94105.

Teacher	Core	Cohort	Labels	Updates	Parameters	Pass
Luby	E1	Historical	999	500	16035	5/5
Luby	DeepSets	Historical	3000	500	17907	5/5
Luby	Transformer	Historical	999	500	16995	5/5
Johansson	E1	Historical	999	500	16035	5/5
Johansson	DeepSets	Historical	999	500	17907	5/5
Johansson	Transformer	Historical	999	500	16995	5/5
ColorTrials	E1	New CF	70000	4000	19396	5/5
ColorTrials	DeepSets	New CF	70000	4000	21268	5/5
ColorTrials	Transformer	New CF	70000	4000	20356	5/5
JP	E1	Historical	999	500	19396	5/5
JP	DeepSets	Historical	999	500	21268	5/5
JP	Transformer	Historical	999	500	20356	5/5
BE base	E1	New CF	6776	4000	19396	5/5
BE base	DeepSets	New CF	6776	4000	21268	5/5
BE base	Transformer	New CF	6776	4000	20356	5/5
GM	E1	New CF	6104–6120	500	19396	5/5
GM	DeepSets	New CF	6104–6120	500	21268	5/5
GM	Transformer	New CF	6104–6120	500	20356	5/5
Greedy	E1	Historical	999	500	16035	5/5
Greedy	DeepSets	Historical	999	500	17907	5/5
Greedy	Transformer	Historical	999	500	16995	5/5

Historical sources are `results/sample-complexity/certification.json`, frozen candidates and raw final rows. New sources are `results/multiteacher-counterfactual/plans/successful-policies.json` and confirmation freezes/rows. Prefix budgets reuse labels and seeds; they are not independent replications or theoretical sample-complexity scaling. The new packages do not rewrite historical noncertifications as successes.

Source commits include Pedro audit `6836d855`, paper study `b5427faf`, integration `a54a18e3`, sample complexity `61f39760`, counterfactual `291d4b82`, degree `71153993`. These are branches/worktrees in shared history, not independent copies of every experiment. Old rows/protocols remain immutable; the previous manuscript is archived under `paper/ucgnn/archive/pre-closure-71153993.tex/pdf`.

Optimizer controls and costs. Independent E1 confirmation compares 500/1,500 updates at nominal 3,000 native labels for Luby, JP and BE, five paired seeds and new final tapes. Only optimizer effort differs along the same training lineage. Six arms and all seed outcomes remain in `results/sample-complexity/optimization-confirmation/results.json`. It identifies additional-optimization effects for those bank/seed conditions, not CF effects or optimal/monotone update counts. Luby and JP complete 1,000/1,000 per seed at both milestones. BE moves from 72,90,84,92,42 successes per 1,000 to 1,000 in every seed; these are the strongest resource-controlled optimizer results in the present evidence. The small-width and no-communication baselines below share native prefixes, 500 updates and paired seed conventions with E1. They are development controls, not optimized competitors or proof that every coloring rule needs communication. No-communication erases received states, relations and masks while retaining local inputs and external permissions. Small uses widths 8/4. Dataset manifests record

Teacher	E1	Small	No communication
Luby	28, 28, 28, 28, 28	0, 0, 28, 1, 0	0, 0, 0, 0, 0
JP	28, 28, 28, 28, 28	10, 0, 15, 0, 0	0, 0, 0, 0, 0
BE base	8, 5, 5, 6, 3	0, 0, 0, 0, 0	0, 0, 0, 0, 0

Table 7: Native development controls at nominal 3,000 labels and 500 updates. Each entry lists successes out of 28 for seeds 83101–83105 individually.

simulation steps, native/added labels, generation CPU/wall time and bytes. Four labels per color root are charged even if queries coincide: these are conservative acquisition counts, not distinct input counts. Checkpoints retain Adam, all RNG states, visitation and query caches. Closure trains no scientific models. Disposable technical gradient tests produce no weights used in results. The evidence ledger identifies all selected hashes and full label-generation costs.

Teacher	Native labels	Added CF labels	Bank CPU seconds	Stored MB
BE base	3000	3776	0.435–0.592	0.874–0.876
ColorTrials	30000	40000	2.153–2.215	1.172–1.186
GM	3000	3104–3120	0.857–0.929	2.002–2.091

Table 8: Acquisition cost of the complete shared native/CF bank, range over five seeds. CPU time includes simulation, encoding and both label types; it is not the isolated marginal CF cost. Decimal MB includes replay witnesses. Training costs are separate.

D Continuation Diagnostics and Dependence

The sufficient key comes from actual BE second-stage inputs, not its label. Real centers have pending bit one and own-slot bits zero; ID/slot residuals are zero. KW substitutes receiver-relative option equality for neighbor own bits. QueryPolicy signs the offset. The root queried slot has offer bit one; other slots vanish under gather. Each valid neighbor contributes occupied query bit, participation and offer/query equality; MAX removes duplicates. With phase, root permission, signed offset and root participation, this determines

	Correct / 99,840				Unseen-type correct		
Seed	Passive	Matched	CF	Subset size	Passive	Matched	CF
94101	41,591	9,343	99,840	549	229	158	549
94102	9,157	9,322	99,818	459	270	342	437
94103	9,146	9,146	99,840	717	229	229	717
94104	42,262	9,076	99,828	522	288	182	510
94105	10,865	9,382	99,840	722	356	246	722

Table 9: Exact BE/E1 original numerators and posthoc novelty subsets. Subset sizes count correlated query occurrences, not distinct types.

the real queried-slot logit at fixed weights. It is not a sufficient key for proposals. Train keys include native and all stored CF queries. There are 58–65 distinct training keys and 115–132 diagnostic keys per seed, with highly unequal occurrence counts. Complete serialized input hashes retain irrelevant fields and measure a different novelty notion.

The diagnostic reconstruction is `scripts/final-counterfactual-audit.py`. It reuses the original queries/RNG, reproduces all numerators, and records each checkpoint and bank hash. Stored label checks find zero inconsistencies; they validate the stated oracle, not graph-wide realizability of every query. We use five paired seed ranges, not decision-level significance. Even five same-sign pairs give exact two-sided sign-test $p = 2/2^5 = 0.0625$ under an exchangeable-sign null, before multiplicity. Shared BE/GM heads and repeated states cannot increase the independent training sample size. Positive counts across seeds are 3,070,2,895,2,891,2,773,3,115; full confusion/eligibility strata are in `results/final-closure/review-ml-be-strata.json`. For unseen-type subsets positives are 52,108,90,56,60 and negatives 497,351,627,466,662. All arms have zero false negatives. The distinction concerns rejecting negative continuations under the explicitly chosen oracle; native traces alone do not uniquely specify off-policy semantics.

E Failure Inventory and Statistical Units

The complete table is `results/final-closure/generalization-cells.csv`. The factor table separates teacher, architecture, checkpoint, seed, size, degree, palette, family and cohort within stage. The following lists all failed confirmatory cells, each with 1,000 runs and $n = 64$. `failed-cells.json` and

Teacher/core	Cohort	Seed	Δ	Successes	Palette
Johansson/Transformer	Historical primary	83101	10	972	Global
GM/Transformer	CF primary	94102	4	819	Local
CT/E1	Native matched	94101	3	999	Global
CT/E1	Native matched	94101	4	999	Global
CT/E1	Native matched	94101	4	999	Local
CT/E1	Native matched	94105	3	994	Global
CT/E1	Native matched	94105	4	987	Global
CT/E1	Native matched	94105	10	912	Global
CT/E1	Native matched	94105	4	993	Local
GM/Transformer	Native matched	94102	4	819	Local
GM/Transformer	Native passive	94102	4	819	Local
CT/E1	Equal-update matched	94104	3	864	Global
CT/E1	Equal-update matched	94104	4	855	Global
CT/E1	Equal-update matched	94104	10	547	Global
CT/E1	Equal-update matched	94104	4	835	Local

Table 10: All fifteen failed cells. CT denotes ColorTrials; equal-update controls are not retrospectively assumed native certified.

`failure-replay-audit.json` give full checkpoint/row hashes, graph/tape seeds, first relevant differences

and trace paths. All fifteen first failures reproduce with original batch context. Terminal auditing reconstructs graphs/fingerprints, conflicts, incompleteness and palette violations and checks stored invariant/failure fields. It does not pretend to reconstruct unrecorded intermediate states for all runs.

Graphs/tapes repeat across cores, teachers, arms and global/local lists. Each checkpoint's instance-index battery supports its own pointwise inference; pooling dependent observations across policies does not. Realized family counts vary under iid uniform family selection. No aggregate defines a sampled population of trained algorithms.

F Output Laws and Preserved Distribution Studies

For fixed labeled G, let $\mu_{\pi,G}$ be the full terminal-vector/status law including failures/cutoffs. Labels, palette, interpreter, cutoff, conditioning and color-name quotient must agree before comparing studies. Equal vertex marginals do not determine a joint law: equally weighted $(0,0)/(1,1)$ versus $(0,1)/(1,0)$ have equal marginals and disjoint support. This logical example is not a proper-coloring counterexample on an edge.

Proposition 8 (Symmetry and coupling)**.** *On a complete graph with exactly n colors, a vertex-equivariant stochastic protocol with exchangeable private inputs is uniform over proper labeled colorings conditional on successful termination of positive probability. Any coupling (X, Y) satisfies $d_{\mathrm{TV}}(\mathcal{L}(X), \mathcal{L}(Y)) \le \Pr(X \neq Y)$.*

Proof. Vertex permutations act transitively on the $n!$ proper coloring bijections; equivariance and the invariant success event make their conditional masses equal. For any event A, the difference in probabilities of $X, Y \in A$ is at most $\Pr(X \neq Y)$; take the supremum. □

Anonymity alone does not establish the symmetry premises. Untransformed IDs, non-equivariant schedules and numerical ordering require care. With more than n colors, vertex symmetry need not equate different color subsets. For r independent paired runs of one graph/policy protocol with zero disagreements, the one-sided upper limit $1 - \alpha^{1/r}$ bounds disagreement and hence TV under that coupling. It says nothing universal about other graphs/checkpoints. Uniform conditional full-transition coupling error ϵ for H common pre-divergence steps gives $d_{\mathrm{TV}} \le H\epsilon$; per-node error would also require a node union bound. Native accuracy is not this uniform premise.

Global-law, local-palette, robust-statistics and factorial studies remain in their historical directories with schemas, fixed banks and testing families. Complete runs are observations, permutations act on rows rather than independently on vertices, and witness selection needs an independent split or valid joint permutation scheme (Gretton et al., 2012; Hemerik and Goeman, 2018; Phipson and Smyth, 2010; Holm, 1979). Reused banks are historical reanalysis, not independent new confirmation. Nonrejection does not establish equal laws or tolerance equivalence; power against localized alternatives matters. These studies support scoped distributional distinctions, not new MMD/permutation theory, and are outside the 810,000 denominator.

G Novelty Comparison and Reproduction Map

Candidate	Established precedent	Defensible increment
Neural coloring	Lemos et al. (2019); Langedal and Manne (2025); Vanderbush and Weber (2026); Le et al. (2026)	Seven-rule native coverage with declared information and complete selected-checkpoint provenance
Trace imitation; MAX	Veličković et al. (2020, 2022)	Concrete validity/native/continuation comparisons for these teachers
Interactive/CF learning	Ross et al. (2011); de Haan et al. (2019); Bevilacqua et al. (2023)	Specific admissibility contracts and paired continuation gaps, not a new general causal method or equal-label advantage

Candidate	Established precedent	Defensible increment
Output versus trajectory	Rodionov and Prokhorenkova (2023); Veličković et al. (2022)	Quantified separations in these coloring interfaces, not novelty of the conceptual distinction
Information/locality	Linial (1992); Loukas (2020); Sato et al. (2019)	JP representation control and interpreter accounting; the elementary observation lemma is not a new lower bound
Algorithm extrapolation	Xu et al. (2021); Mathys et al. (2023); Wittig et al. (2026)	Frozen selected-condition outcomes and replayable palette/preference failures, not a general extrapolation theorem
Randomized output laws	Gretton et al. (2012); Hemerik and Goeman (2018); Holm (1979)	Matched statistical objects and protocol-specific evidence, not new test theory

The focused primary-source search was executed on 26 October 2026, including recent 2026 work. It is not proof of absence of every related study. `docs/NOVELTY_AND_RELATED_WORK.md` records scope and objections. `docs/FINAL_EVIDENCE_AUDIT.md` and `results/final-closure/evidence-register.json` link central claims to proofs, protocols, data, weights, seeds, hashes, limitations and confidence. Run from the repository root without scientific training:

```
PYTHONPATH=src python scripts/final-evidence-audit.py
PYTHONPATH=src python scripts/final-failure-replays.py
PYTHONPATH=src python scripts/final-counterfactual-audit.py
PYTHONPATH=src python scripts/verify-sample-complexity.py -runtime
PYTHONPATH=src python scripts/verify-multicf.py
PYTHONPATH=src python scripts/build-closure-assets.py
PYTHONPATH=src python scripts/build-closure-paper.py
```

The build receipt checks references, figure/table inclusion and PDF text and records compiler/environment hashes and renders. Source-only Git checkouts are not self-contained: large raw/weight artifacts require authenticated runtime archives. Local inventory/recovery commands identify them. Missing Pedro weights are not imputed. Closure does not publish a release, update models or select checkpoints using transfer/final outcomes.

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